SOORAJ KK

Data Engineer | GenAI

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Professional Summary

Data Professional with 2+ years of experience in developing web scraping, data extraction, ETL pipelines, and data processing workflows using Python, SQL, Scrapy, PySpark, and PostgreSQL. Experienced in processing millions of records monthly, integrating data from multiple external sources, and transforming raw data into structured, reliable datasets for analytics and business intelligence. Hands-on experience with AWS, Docker, REST API development, and automated data workflows, complemented by independent project experience in Generative AI, LLM integration, tool calling, and agentic AI applications.

Technical skills

Programming
Python
SQL

Data Engineering
Scrapy
PySpark
Apache Airflow
dbt

Backend Development
Django REST Framework
FastAPI
REST APIs

DevOps & Version Control
Docker
Git
GitHub Actions

Analytics & BI
Microsoft Power BI
Exploratory Data Analysis (EDA)

Databases & Cloud
PostgreSQL
MySQL
AWS S3
AWS EC2

GenAI & Agentic AI
LangChain
LangGraph
Ollama
LLM Tool Calling

Professional Experience

Data Analyst | Orbio Solutions Pvt Ltd 07/2024 – Present | Calicut, India

- Developed and maintained data extraction and ETL workflows to collect, process, and standardize product information from multiple external sources.
- Implemented data cleaning, transformation, validation, and quality checks to improve data consistency and reliability for downstream analytics.
- Automated recurring data processing tasks and troubleshot web scraping failures and pipeline issues to improve workflow reliability.
- Contributed to the development of a retail price intelligence platform for competitor price tracking, historical SKU-level pricing analysis, and cross-platform product comparison.
- Developed backend components and REST APIs to support consumer and distributor dashboards, enabling data access across application components.
- Supported database operations, backend debugging, and performance optimization to maintain reliable data processing and application functionality.
- Investigated and resolved data quality issues across multiple sources to maintain standardized product datasets and support downstream reporting and analytics.

Recent Production Projects

Ecomedes, *Orbio solutions*
Developed scalable batch data extraction and ETL workflows for Ecomedes, collecting sustainable building product data from 25+ external sources. Implemented web scraping, data transformation, and validation processes to deliver standardized product datasets, certifications, and metadata for downstream analytics and reporting.

Features Implemented

- Processed 1–3 million records per month using Scrapy, PySpark, SQL, and PostgreSQL.
- Developed Scrapy spiders to extract product specifications, certifications, and metadata.
- Implemented PySpark workflows for data cleaning, transformation, and validation, storing raw data in PostgreSQL.
- Delivered processed datasets, product certifications, and images to client AWS S3 buckets for downstream analytics and reporting.
- Orchestrated reliable batch pipelines using Apache Airflow with monitoring and retry mechanisms.
- Developed Power BI dashboards to analyze product trends, certification status, and key business metrics.

OttixHow - AI Retail Price Intelligence, *Orbio solutions*

Developed a retail price intelligence platform to collect competitor product data, track historical SKU-level pricing, and support cross-platform product comparison using web-scraped data and LLM-assisted product identification.

Features Implemented

- Developed REST APIs using Django REST Framework to support consumer and distributor dashboard functionality.
- Built Scrapy spiders with proxy rotation and automated retry handling to collect product details and pricing data from multiple e-commerce platforms.
- Designed PostgreSQL schemas to store product information, multi-currency prices, and historical SKU-level pricing data for competitor analysis.
- Explored LLM integration to interpret user product comparison requests and identify matching products in the scraped product database, helping map natural-language queries to relevant product records across platforms.
- Containerized data-processing pipelines using Docker and integrated automated checks into GitHub Actions to support code quality and reliability.

Independent Technical Projects

AI Trip Planner Agent

Developed an AI-powered travel planning assistant using LangGraph and a locally hosted Qwen3:4b model through Ollama. The application uses a tool-calling workflow to generate personalized travel itineraries and integrate external APIs for destination research, weather information, currency conversion, and expense estimation

Features Implemented

- Built an agentic workflow using LangGraph and LangChain to orchestrate LLM responses and tool execution.
- Integrated Google Places and Tavily Search for destination research, attractions, restaurants, activities, and transportation information.
- Integrated OpenWeatherMap for weather information and ExchangeRate-API for currency conversion.
- Implemented expense calculation tools for accommodation costs, total trip expenses, and daily budget estimation.
- Developed a FastAPI REST API for processing travel-planning queries and returning generated responses.
- Built a Streamlit frontend for interactive user input and Markdown-formatted itinerary display.
- Configured a local LLM using Ollama (qwen3:4b) to support local inference without relying on a hosted LLM API.

Education

B.Sc. in Botany, *Calicut University* 07/2016 – 05/2019 | Kerala, India

Certificates

- Data Science 101 (DS0101EN) — IBM SkillsBuild
- Ethics in AI and Data Science (LFS112) — The Linux Foundation
- Data Literacy — IBM SkillsBuild